

Employer's Requirements (Extract)

Nandira River Bridge Project — Package NRB-4

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From:	National Highways Infrastructure Authority (Employer)
To:	Meridian Engineering Consultants (Engineer); Sagara Constructions Pvt. Ltd. (Contractor)

This document sets out the Employer's technical requirements for the design (to the extent stated), construction, testing and handover of the Works. It is a technical requirements document and does not form the Conditions of Contract; where a technical matter set out below has contractual consequence, that consequence is dealt with in the General Conditions or the Particular Conditions, and this document is to be read together with those documents and with the Drawings and Specification. This is an extract; matters not addressed below are not to be taken as excluded from the Contractor's obligations under the Contract.

1. PROJECT OVERVIEW

The Works comprise the design (to the extent stated in this document), construction and commissioning of a new river bridge and associated approach works carrying the National Highway NH-14 realignment across the Nandira River, together with all ancillary civil, structural and drainage works necessary for the bridge and approaches to be brought into safe public use. The Works are located between approximate chainage km 212+400 and km 213+040, Suvarna District, Republic of Vantara.

The Contractor is responsible for constructing the Works in accordance with the reference design issued by the Engineer, this document, the Specification and the Drawings, and for such elements of design as are expressly identified in this document as the Contractor's responsibility.

2. SCOPE OF WORKS

The Scope of Works includes, without limitation: site clearance and preparatory works; temporary diversion of traffic and construction of any temporary access required for construction; foundation and substructure works for the bridge, including piling, pile caps, piers and abutments; superstructure works, including precast or cast-in-situ girders, deck slab, parapets, expansion joints, bearings and drainage; approach road formation, pavement and associated drainage on both banks within the defined limits of the Site; permanent traffic signage and road marking within the limits of the Works; testing, commissioning and load testing of the completed structure; and remedying of defects during the Defects Notification Period.

The Scope of Works excludes permanent electrical street lighting installations, which are the subject of a separate package, save that the Contractor shall provide such conduit and duct crossings within the Works as are shown on the Drawings to accommodate that separate package.

3. BRIDGE CONFIGURATION

The bridge shall comprise 8 spans with an overall length of approximately 640 metres, supported on 2 abutments (Abutment A1 on the left bank and Abutment A2 on the right bank) and 7 piers (Piers P1 to P7, numbered sequentially from Abutment A1), with an average span length of approximately 80 metres. Piers

P3 to P7 are located within or adjacent to the active river channel; Piers P1 and P2 are located on the left bank approach outside the normal river channel.

The bridge shall carry a dual two-lane carriageway with hard shoulders, with an overall deck width of not less than 14 metres, together with parapets and a maintenance walkway on one side as shown on the Drawings. The soffit of the superstructure shall provide a clearance of not less than 6 metres above the Highest Flood Level notified by the Suvarna District Water Resources Department, measured at the deepest point of the river channel beneath each span.

4. STRUCTURAL PERFORMANCE REQUIREMENTS

The Works shall be designed and constructed for a design life of 100 years. Structural design, to the extent it is the Contractor's responsibility under this document, shall be carried out in accordance with the Vantara National Bridge Design Code and such other codes and standards as are identified in the Specification. The bridge shall be designed for the seismic zone and live load class stated in the Specification, and for hydraulic loading corresponding to the design flood discharge stated in the Specification.

Save as expressly stated in this document, the design of the Permanent Works, including the design of the substructure and superstructure, is provided by the Employer through the Engineer, and the Contractor shall construct the Permanent Works strictly in accordance with the Drawings issued for construction, save to the extent an instruction under Clause 3 or Clause 13 of the General Conditions directs otherwise.

5. MATERIALS

Concrete used in substructure elements, including pile caps, piers and abutments, shall achieve a minimum characteristic compressive strength of Grade M40. Concrete used in superstructure elements, including girders and deck slab, shall achieve a minimum characteristic compressive strength of Grade M45. Reinforcement shall be high yield strength deformed bars of the grade stated in the Specification, and shall be accompanied by mill test certificates traceable to the batch of steel used in each pour. Cement, aggregates, admixtures and any precast elements shall conform to the standards and source approvals stated in the Specification, and the Contractor shall obtain the Engineer's approval of each proposed source of materials before that source is used on the Works.

6. FOUNDATIONS

Pier and abutment foundations shall be piled foundations, of the type, diameter and founding criteria shown on the Drawings, founded on strata meeting the bearing and settlement criteria stated in the Specification. Pile installation shall be carried out in accordance with an approved method statement under Section 16 of this document, and each pile shall be subject to the integrity and load testing regime stated in the Specification. Where ground conditions encountered during piling differ from those indicated in the geotechnical information provided with the tender documents, the Contractor shall notify the Engineer promptly and shall not proceed with the affected foundation until the Engineer has reviewed the matter.

7. TEMPORARY WORKS

The Contractor is responsible for the design, adequacy and safety of all Temporary Works, including but not limited to cofferdams, falsework, formwork, staging, access trestles and any temporary works platform required for construction within the river channel. Temporary Works design shall be prepared by a suitably qualified person and shall be submitted to the Engineer for review under Section 16 of this document before

the relevant Temporary Works are constructed. Engineer's review of Temporary Works design does not relieve the Contractor of responsibility for its adequacy.

8. UTILITIES

The Drawings show the location of utilities known to the Employer at the date of issue, including an overhead transmission line owned by Vantara Power Grid Corporation running generally parallel to the highway corridor and crossing it at one or more locations shown on the Utility Drawings, and an underground water trunk main owned by Suvarna Water Board crossing the river near the left bank approach. The Contractor shall verify the location of any utility shown on the Drawings by trial pit or other non-destructive method before carrying out excavation within 5 metres of its indicated position, and shall coordinate the timing of its operations with the relevant utility owner in accordance with Part C.4 of the Particular Conditions.

The utility information provided with the tender documents is based on records supplied by the relevant utility owners and may not be complete or fully accurate. The Contractor shall exercise reasonable skill and care in locating utilities and shall report to the Engineer without delay any utility, apparatus or installation encountered that is not shown on the Drawings or otherwise disclosed in writing before the Commencement Date.

9. ENVIRONMENTAL PROTECTION

The Contractor shall carry out the Works in compliance with the environmental clearance issued by the Vantara Environmental Protection Board and the requirements of Part C.3 of the Particular Conditions. Without limiting those requirements, the Contractor shall prepare and submit to the Engineer, before commencing any in-river work, a River Works Environmental Management Plan addressing turbidity control, spill prevention and response, disposal of spoil and construction waste, and protection of riverbank vegetation outside the limits of the Works. In-river work shall not commence until the Engineer has confirmed no objection to the Management Plan.

10. QUALITY ASSURANCE

The Contractor shall implement and maintain a quality management system appropriate to the nature and scale of the Works, and shall submit an Inspection and Test Plan for each principal element of the Works (piling, substructure concrete, superstructure concrete, and pavement) for the Engineer's review before that element of work commences. Each Inspection and Test Plan shall identify the Hold Points applicable to that element in accordance with Section 11 of this document, the inspections and tests to be carried out, and the records to be generated.

11. INSPECTION AND HOLD POINTS

The following stages of the Works are designated Hold Points, at which the Contractor shall not proceed without the prior written consent of the Engineer or the Resident Engineer, requested in accordance with Part C.6 of the Particular Conditions:

- (a) founding level and pile-cap excavation, before placing blinding concrete;
- (b) reinforcement fixing for any substructure or superstructure element, before placement of formwork over that reinforcement;
- (c) formwork and falsework for any element, before placing concrete;

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- (d) each concrete pour, which shall be witnessed by the Engineer's Representative or the Resident Engineer or their nominee;
 - (e) backfilling around any foundation or substructure element, before backfill is placed;
 - (f) erection or launching of any girder or precast element; and
 - (g) load testing of the completed structure, before issue of the Taking-Over Certificate.

The Contractor shall maintain the Hold Point register required by Part C.6 of the Particular Conditions and shall not treat a Hold Point as cleared until the Engineer's or Resident Engineer's written consent has been recorded in that register.

12. TESTING REQUIREMENTS

Concrete testing shall be carried out at a minimum frequency of one set of three cubes for every pour, or for every 30 cubic metres placed in a continuous pour exceeding that volume, tested at 7 and 28 days by the independent testing laboratory nominated by the Engineer. Reinforcement shall be tested by mill certificate verification and, where required by the Specification, by independent sample testing. Compacted fill and backfill shall be tested for density and moisture content at the frequency stated in the Specification. Pile integrity and load testing shall be carried out in accordance with Section 6 of this document and the Specification. The completed bridge shall be subject to a static and, where required by the Specification, dynamic load test before the Taking-Over Certificate is issued, the results of which shall be reviewed and accepted by the Engineer.

13. SURVEY REQUIREMENTS

The Contractor shall establish and maintain survey control points referenced to the benchmarks and coordinate system stated in the Specification, and shall engage a qualified surveyor to verify the setting-out of foundations, piers and abutments before construction of each element and to record as-built levels and positions on completion. Setting-out and as-built survey shall be within the tolerances stated in the Specification, and survey records shall be submitted to the Engineer for each principal element before the corresponding Hold Point under Section 11 is cleared.

14. CONSTRUCTION RECORDS

In addition to the general record-keeping obligations in Part C.7 of the Particular Conditions, the Contractor shall maintain, for each concrete pour, a pour record identifying the date, location, quantity, source and batching details of the concrete placed, the ambient conditions, and the personnel and Engineer's representative present; and for each pile, a pile record identifying installation date, depth, founding stratum, and the results of any integrity or load test carried out on that pile. These records shall be submitted to the Engineer within five working days of the event to which they relate.

15. SHOP DRAWINGS

The Contractor shall prepare and submit shop drawings, including reinforcement bar bending schedules, formwork and falsework drawings, and precast element drawings, for the Engineer's review not less than fourteen days before the drawing is required for construction. The Engineer shall return each shop drawing marked as approved, approved with comments, or rejected, within seven days of receipt. Construction shall not proceed on the basis of a shop drawing that has not been returned marked approved or approved with comments, and where returned with comments, the Contractor shall resubmit incorporating those comments

before proceeding.

16. METHOD STATEMENTS

The Contractor shall submit a method statement for the Engineer's review before commencing each of the following activities: piling; cofferdam construction and dewatering; Temporary Works erection under Section 7 of this document; concrete pours below water level or of a continuous volume exceeding 100 cubic metres; and erection or launching of girders or precast elements. Each method statement shall identify the sequence of work, plant and equipment to be used, temporary works involved, and the safety measures to be adopted. The Engineer's confirmation of no objection to a method statement does not relieve the Contractor of responsibility for the adequacy of the method or the safety of its execution, and the relevant activity shall not commence until that confirmation has been given.

17. AS-BUILT DOCUMENTATION

Before application for the Taking-Over Certificate, the Contractor shall submit as-built drawings for all Permanent Works, incorporating all changes from the issued-for-construction Drawings; operation and maintenance manuals for bearings, expansion joints, drainage systems and any mechanical or electrical equipment forming part of the Works; a consolidated set of construction records compiled under Section 14 of this document; test certificates and survey records compiled under Sections 12 and 13 of this document; and warranty or guarantee documentation for materials and proprietary items incorporated in the Works. The Engineer shall confirm receipt of complete as-built documentation as a condition of issuing the Taking-Over Certificate.

[END OF EMPLOYER'S REQUIREMENTS EXTRACT]
